1. What is the primary focus of Software Engineering (SE) as opposed to traditional programming?

A) Writing code as quickly as possible

B) Building complete, reliable, and maintainable systems

C) Focusing only on the coding and debugging tasks

D) Working individually on small scripts

E) Avoiding any formal processes or documentation

Answer: B) Building complete, reliable, and maintainable systems

2. The Software Crisis of the 1960s-1990s was primarily characterized by:

A) A shortage of skilled programmers

B) An overabundance of efficient software tools

C) Projects that were over budget, missed deadlines, and often failed

D) The widespread adoption of agile methodologies

E) The invention of the first computer

Answer: C) Projects that were over budget, missed deadlines, and often failed

3. The Ariane 5 rocket explosion is a classic case study in software failure primarily due to:

A) A missing hyphen in the code

B) Poor user interface design

C) Reusing code without proper adaptation and testing

D) A virus infecting the launch systems

E) Ethical violations by the programming team

Answer: C) Reusing code without proper adaptation and testing

4. What is the main purpose of the Software Development Life Cycle (SDLC)?

A) To slow down the development process with unnecessary paperwork

B) To act as a roadmap for building software correctly and efficiently

C) To eliminate the need for testing before deployment

D) To focus solely on the implementation (coding) phase

E) To ensure programmers work in isolation

Answer: B) To act as a roadmap for building software correctly and efficiently

5. A Feasibility Study is conducted to answer which fundamental question?

A) How will we code the system?

B) Who will be the project manager?

C) Should we build this system?

D) What programming language should we use?

E) How many developers do we need to hire?

Answer: C) Should we build this system?

6. Which type of feasibility asks, "Can we build it with our current technology and skills?"

A) Economic Feasibility

B) Legal Feasibility

C) Operational Feasibility

D) Schedule Feasibility

E) Technical Feasibility

Answer: E) Technical Feasibility

7. Assessing whether a project complies with data protection laws like GDPR falls under which feasibility?

A) Technical Feasibility

B) Economic Feasibility

C) Legal Feasibility

D) Operational Feasibility

E) Schedule Feasibility

Answer: C) Legal Feasibility

8. The output of the feasibility study is typically a:

A) Fully functional prototype

B) Feasibility Report

C) Software Requirements Specification (SRS)

D) Test case document

E) Deployed system

Answer: B) Feasibility Report

9. Which SDLC phase immediately follows the Feasibility Study?

A) Implementation

B) Testing

C) Requirements Gathering and Analysis

D) Deployment

E) Maintenance

Answer: C) Requirements Gathering and Analysis

10. A Software Requirements Specification (SRS) document serves as a:

A) User manual for the end-users

B) Contract between the client and the development team

C) Marketing brochure for the software

D) Script for unit testing

E) Database schema design

Answer: B) Contract between the client and the development team

11. Which requirement gathering technique involves studying current reports and manuals to understand existing workflows?

A) Brainstorming

B) Prototyping

C) Document Analysis

D) Interviewing

E) Observation

Answer: C) Document Analysis

12. The requirement "The system must allow users to reset their password" is an example of a:

A) Non-Functional Requirement

B) Constraint

C) Functional Requirement

D) Feasibility criterion

E) Test case

Answer: C) Functional Requirement

13. The requirement "The system must respond to user inputs within 2 seconds" is an example of a:

A) Functional Requirement

B) Non-Functional Requirement

C) Legal Feasibility check

D) Constraint

E) Design blueprint

Answer: B) Non-Functional Requirement

14. A rule like "The system must be developed using open-source technologies only" is a:

A) Functional Requirement

B) Non-Functional Requirement

C) Limitation (Constraint)

D) Feasibility Study outcome

E) Testing protocol

Answer: C) Limitation (Constraint)

15. In a university system, who is most likely considered an END USER stakeholder?

A) The Project Sponsor (e.g., University Board)

B) A payment gateway API

C) The Data Protection Authority

D) A student using the portal to check grades

E) The Dean of the Faculty

Answer: D) A student using the portal to check grades

16. What is a key best practice during requirements gathering?

A) Use highly technical jargon to sound professional

B) Involve all key stakeholders early in the process

C) Document assumptions but avoid validating them

D) Ignore non-functional requirements to save time

E) Guess stakeholder needs to expedite the process

Answer: B) Involve all key stakeholders early in the process

17. A common mistake in requirement gathering is:

A) Using clear, simple language

B) Validating requirements with walkthroughs

C) Guessing stakeholder needs without confirmation

D) Maintaining traceability of requirements

E) Involving users from the beginning

Answer: C) Guessing stakeholder needs without confirmation

18. In the SDLC, the phase where the SRS is translated into a blueprint for the system is called:

A) Requirements Gathering

B) System Design

C) Implementation

D) Testing

E) Feasibility Study

Answer: B) System Design

19. Which of these is typically an output of the System Design phase?

A) Feasibility Report

B) Software Requirements Specification (SRS)

C) Database Schema and UI Layouts

D) Fully tested code modules

E) User training manuals

Answer: C) Database Schema and UI Layouts

20. During the Implementation phase, developers:

A) Gather requirements from users

B) Write the actual code based on design documents

C) Create the high-level system architecture

D) Install the software on the production server

E) Fix bugs found after deployment

Answer: B) Write the actual code based on design documents

21. Testing individual modules or functions is known as:

A) Integration Testing

B) System Testing

C) Acceptance Testing

D) Unit Testing

E) Regression Testing

Answer: D) Unit Testing

22. The type of testing where the entire integrated system is tested to evaluate its compliance with specified requirements is called:

A) Unit Testing

B) Integration Testing

C) System Testing

D) Deployment Testing

E) Feasibility Testing

Answer: C) System Testing

23. The phase where the finished software is installed in the real environment for users is:

A) Implementation

B) Testing

C) Deployment

D) Design

E) Maintenance

Answer: C) Deployment

24. Fixing a bug reported by users after the software has been launched is an example of:

A) Adaptive Maintenance

B) Perfective Maintenance

C) Preventive Maintenance

D) Corrective Maintenance

E) Deployment

Answer: D) Corrective Maintenance

25. Updating an application to make it compatible with a new mobile operating system is an example of:

A) Corrective Maintenance

B) Adaptive Maintenance

C) Perfective Maintenance

D) Preventive Maintenance

E) Feasibility Analysis

Answer: B) Adaptive Maintenance

26. Adding a new feature to a system based on user requests after launch is called:

A) Corrective Maintenance

B) Adaptive Maintenance

C) Perfective Maintenance

D) Requirements Gathering

E) System Testing

Answer: C) Perfective Maintenance

27. In software engineering, ethics primarily guide professionals on how to:

A) Write the most complex code possible

B) Design, develop, test, and use software systems responsibly

C) Maximize profit above all else

D) Hide software bugs from users

E) Work only on projects with large budgets

Answer: B) Design, develop, test, and use software systems responsibly

28. Which of the following is a key ethical value in Software Engineering?

A) Obscurity

B) Bias

C) Transparency

D) Secrecy

E) Exclusion

Answer: C) Transparency

29. The ethical principle of protecting users' personal information is known as:

A) Fairness

B) Responsibility

C) Privacy

D) Transparency

E) Feasibility

Answer: C) Privacy

30. Which organization's code of ethics is specifically tailored for ICT professionals in Australia?

A) IEEE

B) ACM

C) BCS

D) ACS

E) IFIP

Answer: D) ACS

31. The IEEE/ACM Joint Task Force created a code specifically for software engineers known as the:

A) Software Engineering Code of Ethics and Professional Practice

B) Universal Programming Conduct Guidelines

C) International Software Development Accord

D) Code of Software Profitability

E) Agile Manifesto

Answer: A) Software Engineering Code of Ethics and Professional Practice

32. A walkthrough in the context of requirements is best described as:

A) A demo of the final product to the client

B) A peer review meeting to check requirements step-by-step

C) A training session for end-users

D) A process for deploying software

E) A method for writing code

Answer: B) A peer review meeting to check requirements step-by-step

33. Which SDLC phase involves creating mockups or sample screens to help users visualize the system?

A) Feasibility Study

B) Prototyping (a technique in Requirements Gathering)

C) System Design

D) Implementation

E) Testing

Answer: B) Prototyping (a technique in Requirements Gathering)

34. The Mariner 1 spacecraft failure is famously attributed to:

A) A hardware malfunction

B) A cosmic ray bit flip

C) A missing hyphen in the code

D) An unethical engineer

E) Lack of financial feasibility

Answer: C) A missing hyphen in the code

35. Which stakeholder is primarily responsible for funding or owning the software system?

A) End User

B) Internal User

C) Regulatory Body

D) Project Sponsor / Client

E) External Partner

Answer: D) Project Sponsor / Client

36. Operational Feasibility is concerned with:

A) The cost vs. benefits of the project

B) Whether the system will work in the real world and be accepted by users

C) The technological capabilities available

D) The project's timeline and deadlines

E) Legal and regulatory compliance

Answer: B) Whether the system will work in the real world and be accepted by users

37. Schedule Feasibility primarily asks:

A) Is it technically possible?

B) Is it legally safe?

C) Can we deliver it on time?

D) Is it worth the cost?

E) Will users accept it?

Answer: C) Can we deliver it on time?

38. The Therac-25 radiation machine incident is a tragic example of failure due to:

A) Poor testing and missing safety checks

B) Using the wrong programming language

C) Lack of financial investment

D) Ethical hacking

E) A natural disaster

Answer: A) Poor testing and missing safety checks

39. Which of the following is NOT a common technique for gathering requirements?

A) Interviews

B) Questionnaires

C) Observations

D) Deployment

E) Brainstorming

Answer: D) Deployment

40. Economic Feasibility analysis determines if:

A) The technology stack is available

B) The benefits of the system outweigh the expenses

C) The project can be completed by the deadline

D) The system complies with the law

E) Users will find the system easy to use

Answer: B) The benefits of the system outweigh the expenses